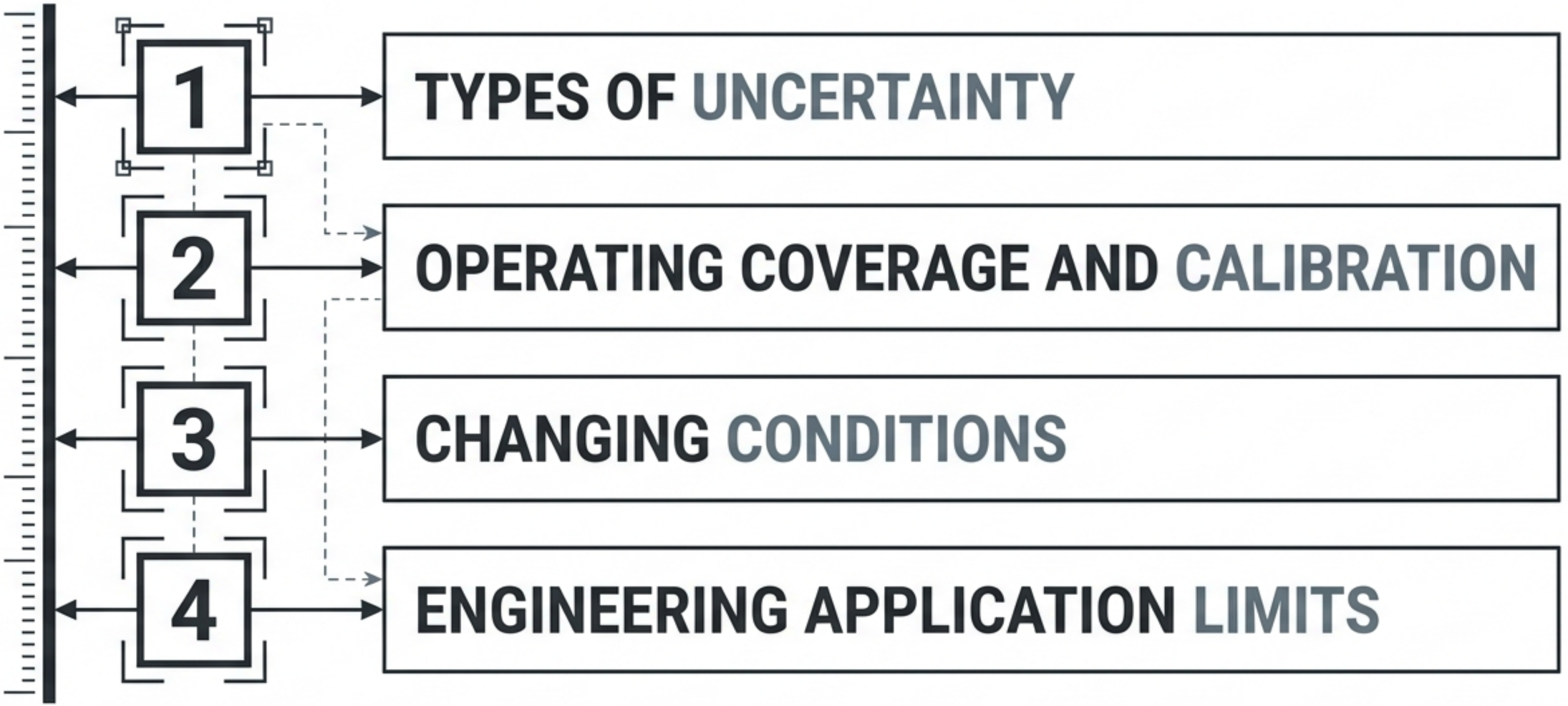
MODULE 15, LESSON 1

AI 101 FOR MECHANICAL ENGINEERS

UNCERTAINTY AND CHANGING CONDITIONS

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COURSE OUTLINE



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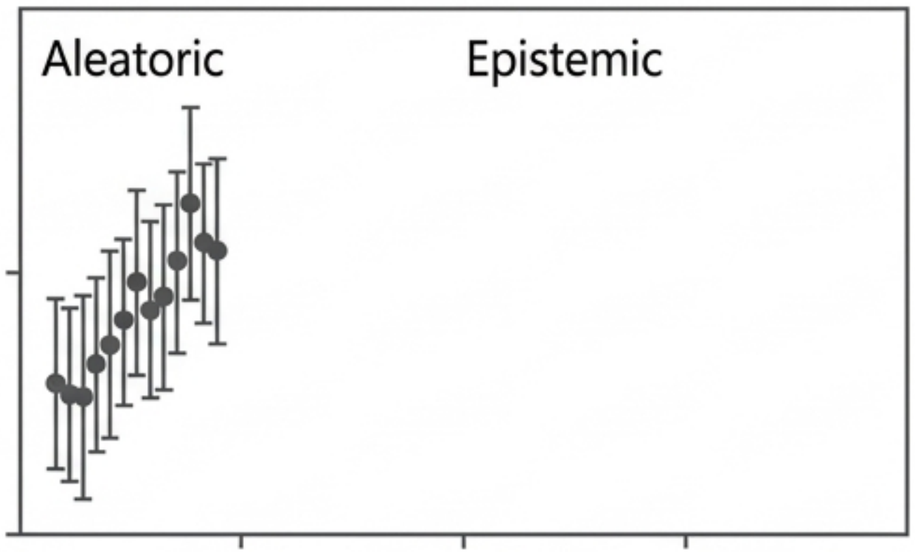
1

Types of Uncertainty

Distinguish inherent observation variation from limits in data and assumptions.

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Classifying Data and Model Uncertainty



Term	Nature	Cause	Mechanical Example
Aleatoric Uncertainty	Unresolved observed variation	Inherent variation	Measurement noise
Epistemic Uncertainty	Limited knowledge or evidence	Data limits or assumptions	Sparse load data

- Aleatoric uncertainty stems from inherent variation in unresolved processes.
- Epistemic uncertainty arises from limited knowledge, sparse data, or assumptions.
- Informative observations can reduce some epistemic uncertainty.
- An inadequate mathematical model will remain fundamentally incorrect.

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2

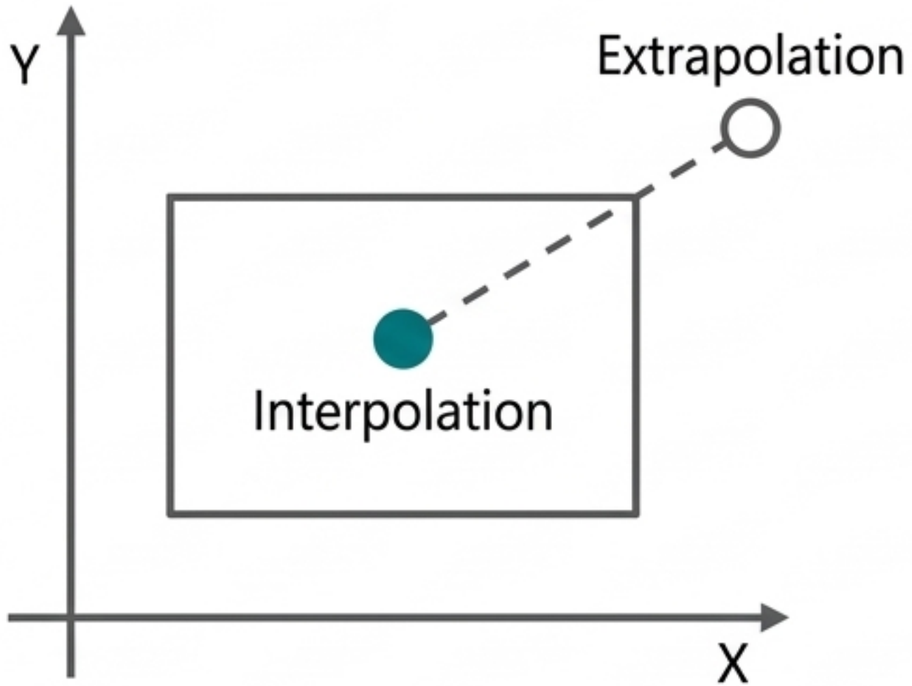
Operating Coverage and Calibration

Define reliable operating boundaries and evaluate statistical probability agreement.

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The Bounded System

Interpolation Versus Extrapolation Constraints

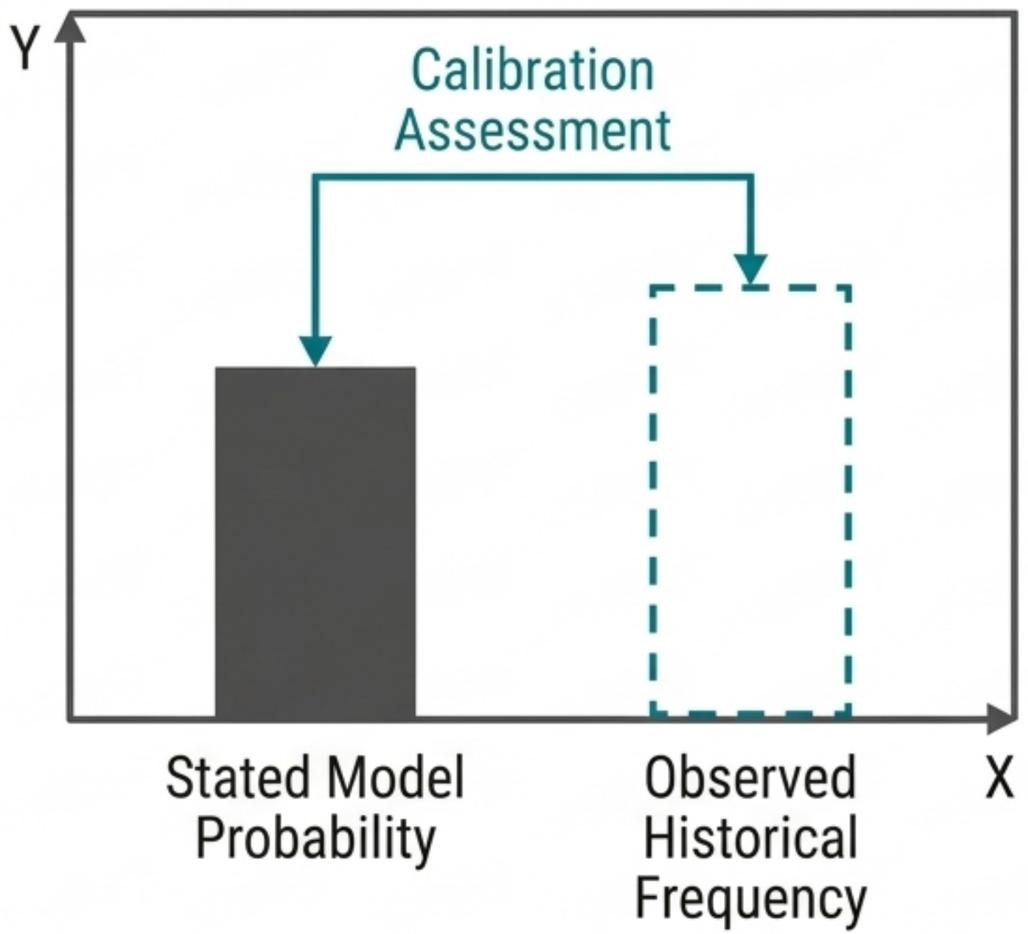


State	Domain Bounds	Input Status	Reliability
Interpolation	Within covered bounds	Sampled operating region	Reliable estimation
Extrapolation	Beyond covered bounds	Unobserved operating regime	Unvalidated estimation

- Interpolation estimates outcomes strictly within covered input conditions.
- Extrapolation estimates outcomes beyond previously covered input conditions.
- Residing inside simple input bounds does not guarantee representative coverage.

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Assessing Probability Calibration



- Calibration compares stated probabilities against historically observed frequencies.
- Assessment must occur under defined, appropriate physical conditions.
- High mathematical model confidence is not automatically calibrated.
- Validate fault probability groups exclusively using independent operational data.

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3
Changing Conditions

Categorise data distribution alterations during real-world operational deployment.

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Distribution Shift and Temporal Drift



Concept	Nature of Change	Temporal Factor	Mechanical Example
Distribution Shift	Deployment differs from development	Discrete event	Plant transfer
Drift	Data relationships alter	Continuous change over time	Sensor wear

- Shift occurs when deployment populations differ from development populations.
- Drift describes gradual changes in data or relationships over time.
- Engineers must continuously monitor both inputs and output outcomes.
- Setting a constant random seed cannot remove distribution shift.

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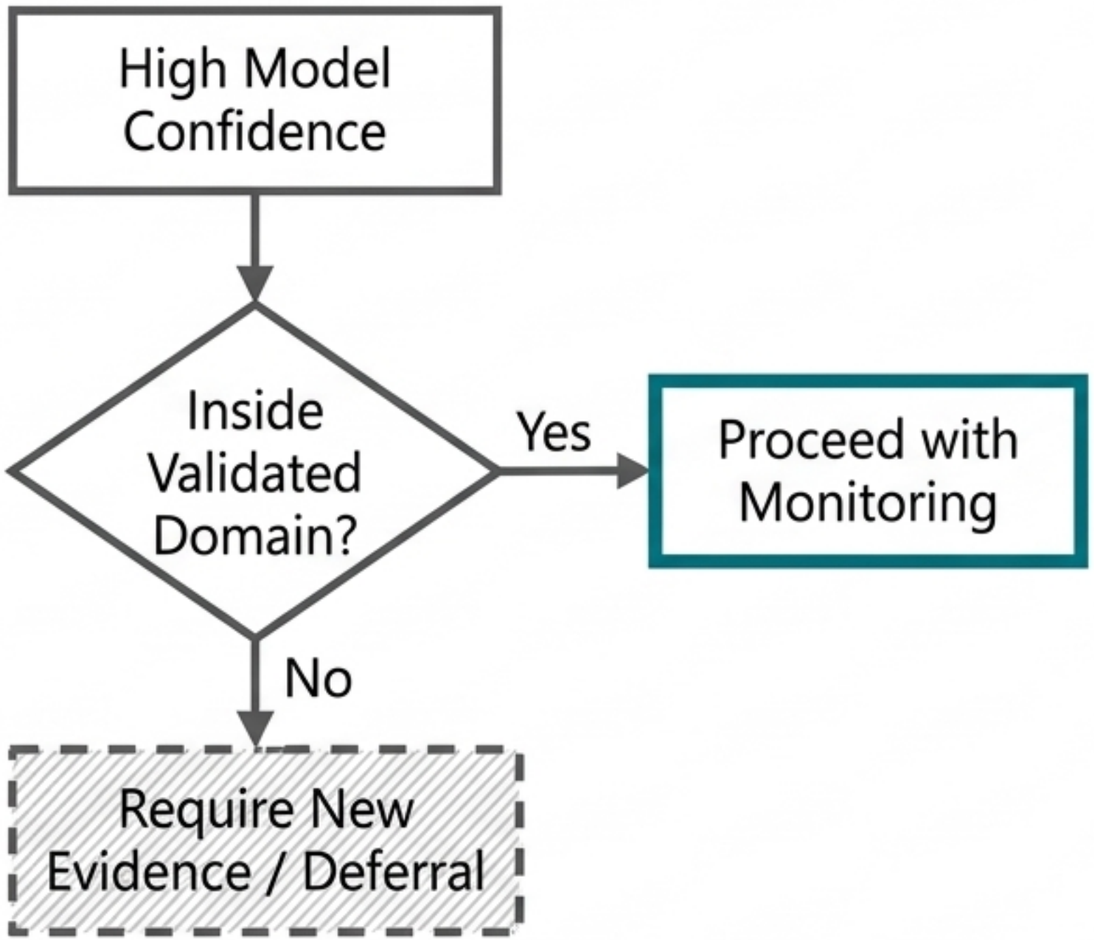
4

Engineering Application Limits

Establish strict rules for model utilisation in dynamic physical environments.

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Constraints on Extrapolated Application



- High model confidence never justifies application outside validated operating domains.
- Confidence metrics degrade rapidly and become poorly calibrated under distribution shift.
- Operating outside evaluated domains strictly requires generating new empirical evidence.
- Unvalidated operational states demand immediate deferral or system retraining.

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Key Points

- **Section 1:** Isolate inherent measurement noise from correctable epistemic data limitations.
- **Section 2:** Constrain models within covered bounds and mandate independent confidence calibration.
- **Section 3:** Monitor systems continuously to detect discrete shifts and temporal drift.
- **Section 4:** High mathematical confidence never overrides the necessity for validated domain evidence.

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